

SECTION DE PHYSIQUE – INTERNSHIP SUBJECT FORM



EPFL SB SPH
PH A2 362 (Bâtiment PH)
Station 3
CH 1015 Lausanne

Téléphone : + 4121 693 5
Fax : + 4121 693 44 44
E-mail : annelene.odegaard@epfl.ch
Site web : <http://sph.epfl.ch/>

Company name and city: Météo-France, Grenoble, France

Project title: Machine learning methods for meteorological time series imputation and forecast: reconstruction of climatological data from observation stations.

Candidate: Léo Catteau

Internship background (max. 200 words):

Climate change since the beginning of the industrial era, due to human activities, is now well documented (IPCC, 2021). In the French mountains, the S2M reanalysis (Vernay et al., 2019), a hybrid dataset combining modelling and observations, and available from 1958 to the present day, is largely used to characterize the meteorological and surface conditions of the last decades.

Nevertheless, it has been shown that temporal heterogeneity in the availability of observations used to correct the model's raw outputs leads to different modelling adjustments over the simulated period (1958-2023) (Vernay et al., 2022). This heterogeneity is most likely the cause of an artefact observed in the temperature trends derived from the reanalysis, leading to cooling over the last 60 years (Vernay et al., 2022), inconsistent with the literature on the subject (Beaumet et al., 2021, Monteiro and Morin, 2023).

Objectives (max. 200 words):

The goal of the internship is to resolve this inconsistency by reconstructing the set of reference observations of past climate directly from data. For this purpose, we will use machine learning methods from multilinear regression baselines to graph neural networks, to reconstruct past missing data from training on recent period. The objective is

to provide usable code implementing different reconstruction methods evaluated on their performances (cost, complexity, reliability of reconstruction...).

Focusing initially on hourly temperatures data, if time permits it will be possible to extend the application of the reconstitution algorithm to other nivo-meteorological variables, or to test the impact of incorporating these reconstructed datasets on temperature trends simulated by reanalysis.

Supervisor's name: Isabelle Gouttevin, Diego Monteiro

Date 19/05/2025

Supervisor's signature:

A handwritten signature in black ink, appearing to be 'DM', written over a horizontal line.

The document can be written in either English or French and must be sent to the internship coordinator (annelene.odegaard@epfl.ch) a few weeks after the internship start.



Internship project

Reconstruction of historical climate time series with graph recurrent neural networks

Author:

Léo Catteau

Directors:

Diego Monteiro

Isabelle Gouttevin

Météo-France, CNRS, CNRM, Centre d'Études de la Neige
Grenoble, France

MSc Applied Physics
École Polytechnique Fédérale de Lausanne
Course 2024-2025

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1 Introduction

Machine learning methods have been playing an increasingly important role in climate sciences, and produced interesting results from problem specific tasks to weather forecasting, raising profound questions about modeling with ongoing efforts to combine traditional large-scale numerical physics models with statistical approaches. When dealing with many unknown physical parameters, or non-physically defined tasks, a possible strategy is to learn the problem directly from data.

Extrapolating missing data is a common task in meteorology, whether it is for treating incomplete field measurements, or for downscaling (for example of outputs from numerical weather prediction systems). As the measurement capabilities augments, especially in the domains of satellite and ground-based remote-sensing, data and physical understanding of meteorological processes progress, and remote regions such as the mountain areas become better monitored. Although, long-term monitoring of such regions are still scarce, due to accessibility issues and harsh conditions for year-round maintenance of instruments. However, climatic studies require long term observations to extract statistically robust characteristics and low frequency signals. One of the key challenges for climate change assessment and understanding in data-scarce regions lies in obtaining the best possible estimation of past climate conditions, which relies both on the few observational data measurements available and on the physical knowledge of climate processes (partly encapsulated in climate models).

The main approach to generate such historical meteorological records is to run large scale physical atmospheric models, constrained on past observations by data assimilation methods. The outcome of such processes is called a reanalysis product, useful to numerous applications, as the production of boundary conditions for application models at the extent of hydrological models. This reanalysis approach is followed by the main meteorological and climate agencies around the world, like NOAA for the United States of America or the European Center for Medium Range Weather Forecasting (ECMWF), with widely used reanalysis products like NCEP/NCAR reanalysis (Kalnay et al., 1996) or ERA5 (Hersbach et al., 2020), respectively. As the main source of continuous meteorological fields data covering large periods, it is at the origin of performances of all subsequent data-based methods, such as praised machine learning forecasting models (most of all trained on ERA5). While reanalyses are successful at providing spatio-temporally (and to a certain extent, physically) consistent data, it has been shown that they can inherit the biases from the structure of the original observations datasets (Vernay et al., 2022). Typically, the further the studied period lies in the past, the fewer observations are available, inducing a temporal heterogeneity in the assimilated dataset with full compartments of the climate system at the global scale almost absent in the most remote data, and hence underrepresented in past data (the mountains can be seen to some extent as one of these compartments).

In this internship, we follow the hypothesis that statistical learning methods can be an interesting approach to solve a combined task-specific (extrapolation of missing values) and physical modeling problem. In the direction of combined efforts of statistical and physical approaches, we propose to use machine learning with deep neural networks to impute missing values in time series of weather fields directly from data, and thus complete observations datasets used to constrain the physical model during a reanalysis process, hopefully correcting the inherited biases in its outcomes.

2 Contributions and related work

Related work: The most elementary imputation techniques are deletion and single imputation methods such as replacement by the mean, to which can be added physical model based approaches for meteorological data (Hamzah et al., 2020). More advanced Bayesian statistical methods have been developed in the context of data assimilation to reconstruct missing signals in multivariate time series from meteorological stations measurements; in a context very similar to the one of our task, only with small datasets and less extreme missing patterns (Chapon, Ouarda, and Hamdi, 2023).

Classical machine learning methods such as clustering can improve performances of elementary mean imputation methods (Rafii and Kechadi, 2019). Some works proposed to use machine learning with neural networks, particularly recurrent neural networks, focusing on the temporal information of a few coupled time series records (Flores, Tito, and Centty, 2020, Huang, Yang, and Fu, 2020).

Deep learning methods have recently been successfully applied to the reconstruction of large scale climate fields, all the more in the context of reconstructing historical fields with large amounts of missing data, focusing on spatial information with Fourier convolution and image inpainting methods (Bochow et al., 2025, Kadow, Hall, and Ulbrich, 2020). These works share our goals of reducing uncertainties and biases in climate records, and have even been applied to improving ocean reanalyses (Lentz et al., 2025).

Contributions : In this project, we propose a deep learning method to reconstruct historical meteorological data with large amount of missing values, at the scale of the French territory. The ultimate goal is to reduce biases in the S2M reanalysis (Vernay et al., 2022) by assimilating reconstructed past meteorological data in addition to the existing historical data. The method is applied at an intermediary scale from previously proposed works: still at large scale (encompassing a large amount of stations), it applies to specific station records, as required by the task and in contrast with large scale fields. This allows accessing both temporal and spatial information through the time series data modality, on the irregular spatial distribution of observation stations. To do so, we rely on a graph representation of data, implementing the Graph Recurrent Imputation Network (GRIN) architecture stemming from recent works in Machine Learning (Cini, Marisca, and Alippi, 2022).

We develop and analyze different training frameworks for the implementation of the GRIN architecture to our reconstruction task. We run different machine learning experiments to set methodological choices, fix hyperparameters and other architecture settings, and adapt the method to our task.

For its evaluation, we develop two different experimental settings: a small scale homogeneous reconstruction task to study fundamental performances without bias, and a heterogeneous reconstruction task over climatological time periods simulating the problem of reconstructing historical data. With the objective of generating physically coherent time series, we develop metrics and losses adapted to the physical/meteorological context of the dataset. With stated best performances, we finally evaluate the climatological coherence of reconstructed time series.

The characterized method achieving satisfactory results, we then produce a meteorological data reconstruction tool with flexible code implementation for an operational application at Météo-France.

3 Data and case study

S2M reanalysis A reanalysis is the product of the simulation of large scale guess (physical model) constrained by field observations that can be re-run over past measurements to reconstruct surface meteorological fields over extended time periods. The frame of this work is the reconstruction of the dataset used to constrain the S2M reanalysis. The S2M reanalysis (Vernay et al., 2022) is a hybrid dataset combining modelisation and observations (Fig.1a), available from 1958 to the present day, and largely used to characterize the meteorological and surface conditions of the last decades over the French high-mountain regions (Alps, Pyrenees and Corsica). It has been shown that temporal heterogeneity in the availability of observations used to correct the model's raw outputs, leads to heterogeneous model adjustments on the course of the simulated period (1958-2023) (Vernay et al., 2022). Indeed, a much larger number of observations is available over the last few decades (1990-2020) compared to the period 1958-1990. The correction applied to the model outputs during the assimilation step hence varies over time, depending on the availability of observations. These variations, though useful in some aspects as bringing the model closer to whatever observation are available, are also most likely the cause of an artifact in the temperature trends derived from the reanalysis. Indeed, this temperature trends feature a cooling over the last 60 years (Fig.1b) that is inconsistent with the literature on the subject, pointing towards an important climate change signal and temperature elevation in the French mountain areas (Beaumet et al., 2021, Monteiro and Morin, 2023). The first objective of this internship project is to produce a reconstructed homogenized observational dataset to feed and constrain the S2M reanalysis, hopefully correcting the targeted bias in trends.

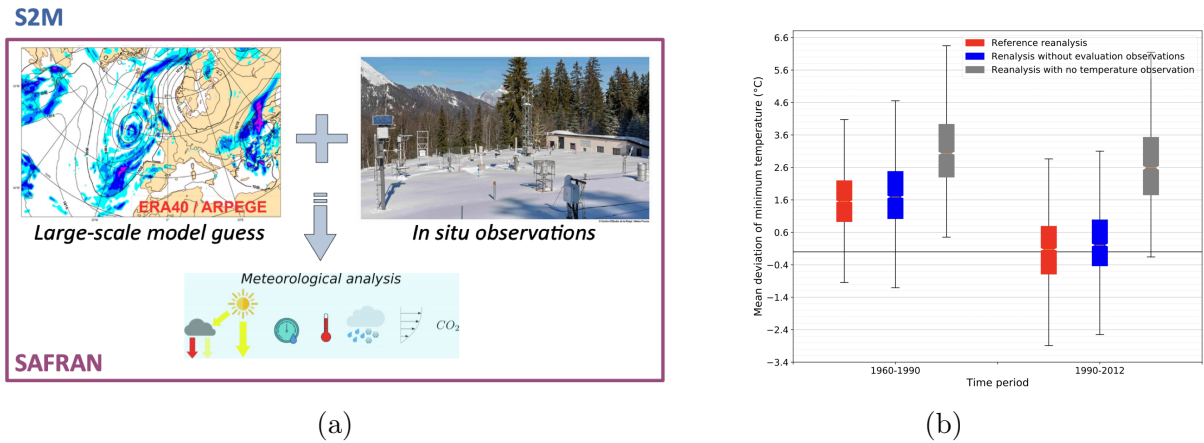


Figure 1: (a) Schematic representation of the S2M reanalysis system. (b) Explicated bias in trends reconstructed by the S2M reanalysis (figure from - Vernay et al., 2022).

Dataset The reconstruction of historical data from available measurements can be treated through the lens of an imputation problem, where a large majority of data is missing in the early part of the dataset. Even if data missing above 50% is usually considered as excessive for statistical methods, we can use model-based and machine learning imputation methods to solve the task, leveraging the large amount of data available in the recent period. Following the framework of time series imputation (Wang et al., 2024, Hamzah et al., 2020), missing patterns in our problem are identified as a mix of random block missing (random start and end of missing patterns) in recent period and monotone block missing (random start and definitive disruption) in past period. The data is considered missing at random (MAR), as the missing process doesn’t depend on the value missing, but not missing completely at random (MCAR) as it depends on the point in time of the observation.

The dataset consists in hourly temperature time series from approximately 2000 stations distributed across mountain areas of the french territory; the area of study of the S2M reanalysis (Fig.2a). Every station is associated to meta-data giving information about geographical position and altitude, but also surrounding topography such as slope at different aggregation scales. Temperature time series range from 1950 to the present day with a large drop in the number of stations available before 1990 to less than 10% of the whole dataset (Fig.2b).

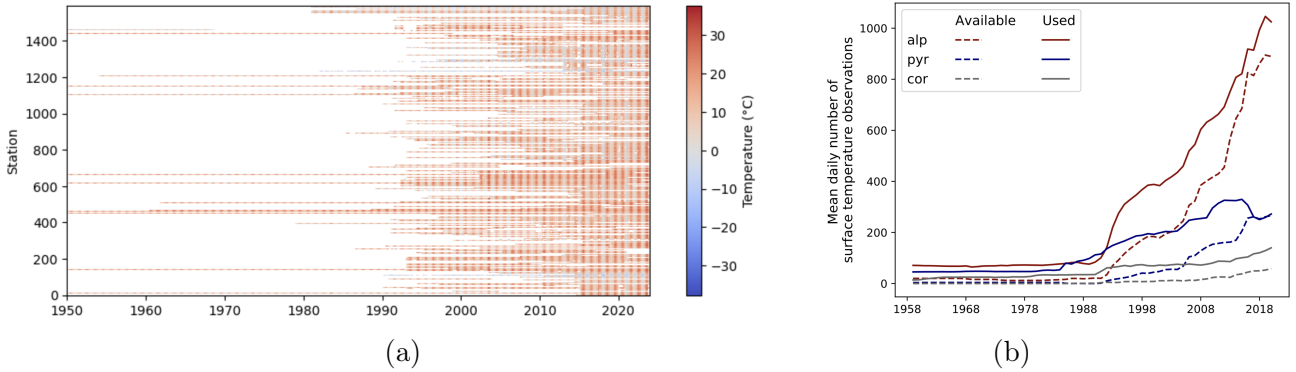


Figure 2: (a) Complete dataset of temperature time series. (b) Number of available observation stations by region as function of time (figure from - Vernay et al., 2022).

4 Methods

We consider the task of reconstructing multivariate time series $X_t \in \mathbb{R}^d$, over a time window $t \in [0, N]$, where at each time step some value x_t^i can be available or missing. We track missing values with a mask $M_t \in \{0, 1\}^{N \times d}$ such that $m_t^i = 0$ if x_t^i is missing and $m_t^i = 1$ otherwise. If $Y_{[t, t+T]}$ are elements of the ground truth complete time series, the objective is to reconstruct complete time series as close as possible to ground truth, from the corrupted ones. In practice, we aim to find the best approximation of the function that would output the reconstructed time series $\hat{X}_{[t, t+T]}$ from $X_{[t, t+T]}$ and $M_{[t, t+T]}$ as:

$$Y_{[t, t+T]} \approx \hat{X}_{[t, t+T]} = f(X_{[t, t+T]}, M_{[t, t+T]}). \quad (1)$$

4.1 Models

Statistical methods The simplest accessible statistical method after deletion is single imputation by the mean, where we replace every missing value by the average of all available values. The average can be taken over the whole dataset, or along a specific dimension as:

$$\hat{X}_t^i = \begin{cases} X_t^i, & \text{for } m_t^i = 1, \\ \frac{1}{d} \sum_{j=1}^d X_t^j, & \text{for } m_t^i = 0. \end{cases} \quad (2)$$

It so happens that the hourly temperature data is very structured temporally, constrained by clear frequency signals of the diurnal and yearly cycles, making it very advantageous to take the average over the spatial variability at each time step, and already allowing to reach relatively low imputation errors. Statistical approaches without learning can be improved in many ways, typically by introducing physical modeling. A simple improvement that can be made from the mean imputation is to introduce a physical continuity prior by weighting the contributions of neighboring stations to the average by their distance to the imputed station as:

$$\hat{X}_t^i = \begin{cases} X_t^i, & \text{for } m_t^i = 1, \\ \sum_{j=1}^d \left(\frac{\frac{1}{x_j^2 + y_j^2}}{\sum_{j=1}^d \frac{1}{x_j^2 + y_j^2}} \right) X_t^j, & \text{for } m_t^i = 0, \end{cases} \quad (3)$$

where geographical coordinates are noted $\{\mathbf{x}, \mathbf{y}\}$ and the sum running over the spatial dimension. Note that the two-dimensions distance used here can be enriched by an altitudinal component to reflect also the vertical distance and the associated disparity between stations. In the following, we call this method geographical coordinates (GC) mean. Other statistical methods without learning can be implemented, such as singular value decomposition to learn principal components of the data, or D-vine copulas sampling from the posterior distribution of a missing value conditional on available observed stations (Chapon, Ouarda, and Hamdi, 2023). In the following, we use mean imputations methods as a statistical lower-bound baseline (intended to be outperformed by more complex methods) as comparison basis for the following machine learning methods.

Multi Layer Perceptron (MLP) Multi Layer Perceptrons are the basic feed forward neural network architecture without (intentional) structure prior. An input vector is treated by N layers composed of linear combinations with trainable weights W and biases b , with a nonlinear activation function σ at every (hidden) step:

$$\text{MLP}(X_t) = W_N \sigma(\dots \sigma(W_2 \sigma(W_1 X_t + b_1) + b_2) \dots) + b_N. \quad (4)$$

The sizes of weight matrices W fix the dimensions of hidden intermediary states. For time series imputation, we feed the corrupted temperatures vector at a given timestep as input and train the network to output a completed imputation vector, from which we keep only the originally missing values giving:

$$\hat{X}_t = M_t \odot X_t + (1 - M_t) \odot \text{MLP}(X_t). \quad (5)$$

Along training, MLP learns the spatial structure of the dataset through the correlations between input stations determining correct missing values. Hopefully, MLP learns the structure in every encountered temperature configuration and is able to extrapolate missing values of the field at different time steps, but only learns with spatial correlations. Because of their flexibility, MLP are used repeatedly inside other architectures as intermediary steps of data processing.

Recurrent Neural Network (RNN) Recurrent Neural Networks are a type of neural network architecture typically suited for data with temporal coherence, where a representation vector h_{t-1} of the history of previous time steps of the time series is fed to the network with the data X_t itself and updated at every step. This creates a connection between (hidden) network units at different time steps, allowing for memory effects and access to temporal correlations (Fig.3a). For the treatment of temporal structure in our imputation task, we consider Gated Recurrent Unit (GRU), a simplified version of the Long Short Term Memory (LSTM) RNNs. In LSTM networks, the representation vector is split into two contributions, supposedly storing short and long term correlations (memory) respectively. In a GRU network, these two contributions are aggregated into one embedding state h_t as the competition between reset and update gates (respectively transforming input data and representation vector itself with previous embeddings):

$$r_t = \sigma(W_{ir}X_t + b_{ir} + W_{kr}h_{t-1} + b_{kr}) \quad (6)$$

$$u_t = \sigma(W_{iu}X_t + b_{iu} + W_{ku}h_{t-1} + b_{kz}) \quad (7)$$

$$\hat{h}_t = \tanh(W_{ih}X_t + b_{ih} + r_t \odot (W_{kh}h_{t-1} + b_{kh})) \quad (8)$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \hat{h}_t. \quad (9)$$

r_t is the reset gate vector, u_t the update gate vector, $\hat{h}_t$ a first step candidate representation vector, h_t the final activation vector fed to next state, X_t the input data and the symbol $\odot$ denote the Hadamard product. In practice, reset and update gates are one-layered MLP, trainable linear projections from input data X_t and previous step embedding h_{t-1} combined with a nonlinearity σ , with trainable parameters W and bias b .

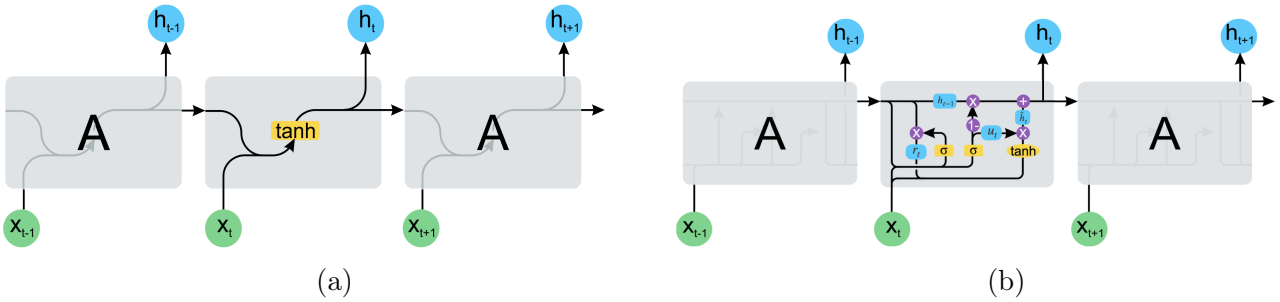


Figure 3: Graphical representations of one cell of (a) Recurrent Neural Network and (b) GRU architectures.

For the imputation task, we use RNN as an encoder, to which we feed the input sequence $X_{[t,t+T]}$ (and the corresponding mask $M_{[t,t+T]}$), and from which we collect back the embeddings at every timestep $h_{[t,t+T]}$. These embeddings represent the states and history of the time series, hopefully in a more expressive way than the temperatures themselves, allowing temporal correlations to be used for imputation. The complete output imputed time series are then obtained with a linear layer decoder that projects back embeddings, from which we keep only missing values:

$$h_{[t,t+T]} = \text{GRU} (X_{[t,t+T]}, M_{[t,t+T]}) , \quad (10)$$

$$\hat{X}_{[t,t+T]} = M_{[t,t+T]} \odot X_{[t,t+T]} + (1 - M_{[t,t+T]}) \odot \text{MLP} (h_{[t,t+T]}) . \quad (11)$$

Graph Neural Network (GNN) Any multivariate time series can be represented as a sequence of weighted graphs, where we consider a graph G_t with d nodes at each time step t (Fig.4). An undirected graph $G_t(X_t, A)$ is defined by a set of nodes $X_t \in \mathbb{R}^d$ and a set of edges encoded in an adjacency matrix A with a_{ij} the weight of the edge connecting the pair of nodes (ij) . Every node can then be identified to one time series at a given time step; the graph based representation enabling the expression of spatial information between time series. The Graph Recurrent Imputation Network (GRIN) is a graph based recurrent architecture taking advantage of the full available spatial and temporal expressiveness of multivariate time series; with RNNs treating in every node the temporal correlations while their spatial interdependence is managed by message passing between nodes over the graph structure. The architecture is composed of two blocks, a spatio-temporal encoder and a spatial decoder, which successively and progressively process the input sequence of graphs.

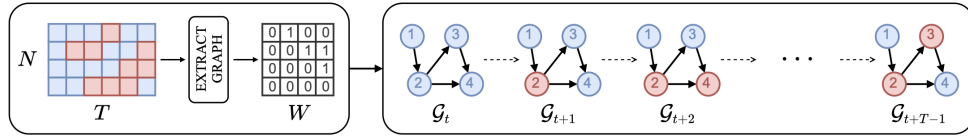


Figure 4: Representation of a multivariate time series as a sequence of graphs. Red circles denote nodes with missing values (figure from - Cini, Marisca, and Alippi, 2022).

In the spatio-temporal encoder, the input sequence $X_{[t,t+T]}$ and mask $M_{[t,t+T]}$ are processed one step at a time, by means of a recurrent neural network GRU cell with gates implemented by message-passing layers. In addition to a usual GRU layer, here the gates and representation vectors are then correlated in their spatial dimension for output by a neural network that works with the graph structure. For it to work well at imputation task, we give as input to the message passing network the concatenation of input data, mask and precedent embeddings. At the node level, the elements of the message-passing GRU (MPGRU) can be described as:

$$r_t = \sigma \left(\text{MPNN} \left(\left[\hat{X}_t^{(2)} \| M_t \| h_{t-1} \right], W_t \right) \right) \quad (12)$$

$$u_t = \sigma \left(\text{MPNN} \left(\left[\hat{X}_t^{(2)} \| M_t \| h_{t-1} \right], W_t \right) \right) \quad (13)$$

$$c_t = \tanh \left(\text{MPNN} \left(\left[\hat{X}_t^{(2)} \| M_t \| r_t \odot h_{t-1} \right], W_t \right) \right) \quad (14)$$

$$h_t = u_t \odot h_{t-1} + (1 - u_t) \odot c_t \quad (15)$$

with r_t , u_t the reset and update gates respectively, h_t the hidden representation of the vector of nodes at time t and $\hat{X}_t^{(2)}$ the output of the decoding block at the previous time-step. The symbol $\|$ denote the concatenation operator. The Message Passing Neural Networks (MPNN) are order k convolutions over the graph, correlating iteratively all neighboring nodes within a distance of k edges. To access the correlations at every edge step, we concatenate the matrix products of input vector of nodes (temperature, mask, latent states) with different powers of the adjacency matrix, and make the readout with a trainable linear layer:

$$\text{MPNN} (X_t, W_t) = \text{MLP} \left(AX_t \| A^2 X_t \| \dots \| A^k X_t, W_t \right). \quad (16)$$

By carrying out the above computation time and node wise to the input X_t , we get the encoded vector H_t . It represent the shared states and history of time series in every node, taking into account its neighborhood.

In the spatial-decoder, a first decoding step is made generating one-step-ahead predictions from the hidden representation $H_{[t,t+T]}$ with a linear readout whose output $\hat{Y}_t^{(1)}$ is used to fill only missing values in X_t as

$$\hat{Y}_t^{(1)} = H_{t-1}W_h + b_h, \quad (17)$$

$$\hat{X}_t^{(1)} = M_t \odot X_t + (1 - M_t) \odot \hat{Y}_t^{(1)}. \quad (18)$$

We then obtain a first complete imputed time series $\hat{X}_t^{(1)}$ from the treatment by message passing recurrent networks. But to take fully into account the spatial structure (more informative in our problem), we update these predictions with the spatial correlations of the time series themselves. The first node-level predictions $\hat{X}_t^{(1)}$ are then concatenated to the mask M_t and the hidden representation H_{t-1} to be processed by a final one-layer MPNN which computes for each node a new (spatial) imputation representation s_t^i as

$$S_t = \text{MPNN}(X_t, W_t) = \text{MLP}(AX_t \| A^2X_t \| \dots \| A^dX_t, W_t), \quad (19)$$

with a convolution that connects this time all the nodes of the graph together. Finally, we generate a linear readout from the the two (spatial and temporal) embeddings that is used to fill missing values in X_t

$$\hat{Y}_t^{(2)} = (S_t \| H_{t-1})W_h + b_h, \quad (20)$$

$$\hat{X}_t^{(2)} = M_t \odot X_t + (1 - M_t) \odot \hat{Y}_t^{(2)}. \quad (21)$$

This is the imputation vector for all the nodes at step t , it is then fed back to the MPGRU cell to process the next input graph G_{t+1} . Applying the previous calculations time and node wise iteratively to the input sequence $X_{[t,t+T]}$, we get the imputed sequence $\hat{X}_{[t,t+T]}$. To be complete, the GRIN architecture accounts for forward and backward dynamics with two encoder-decoder blocks processing time series in two directions, whose output imputations are aggregated by a MLP to generate the final imputation (Fig.5). Given the intrinsically asymmetric masking process of our task, purely backward GRIN block can be advantageously used for historical data reconstruction.

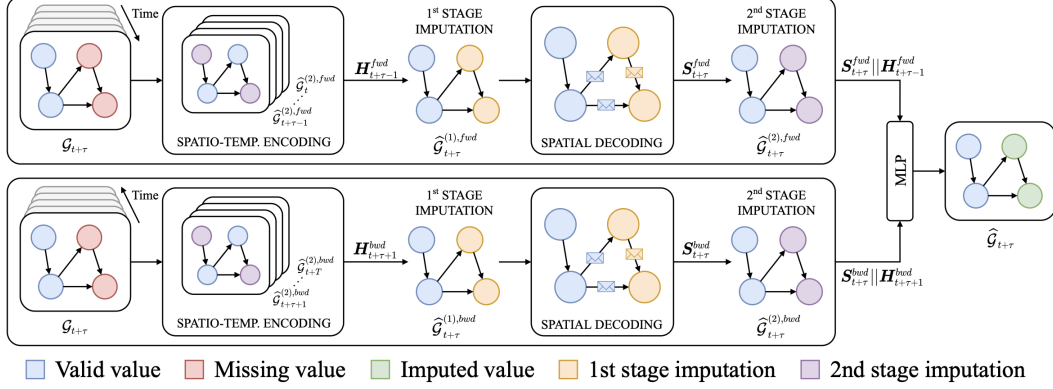


Figure 5: Graphical overview of the bidirectional GRIN architecture (figure from - Cini, Marisca, and Alippi, 2022).

In the underlying graph of the architecture, every node corresponds to one time series (an observations station). In practice, the graph can be constructed directly from data by weighting the edges with the empirical spatial correlation matrix of temperature records. Although, using the exogenous stations meta-data to construct the graph gives relational information with a more pronounced structure, and leads to better performances (see C.2). In the following, we then infer the graph structure by clustering nodes with different distance measures between the stations in the space of exogenous parameters. Close connected nodes then corresponds to the same typology of stations (in terms of geography, topography, ...) as it can be seen with the recovered geographical regions in Fig.6.

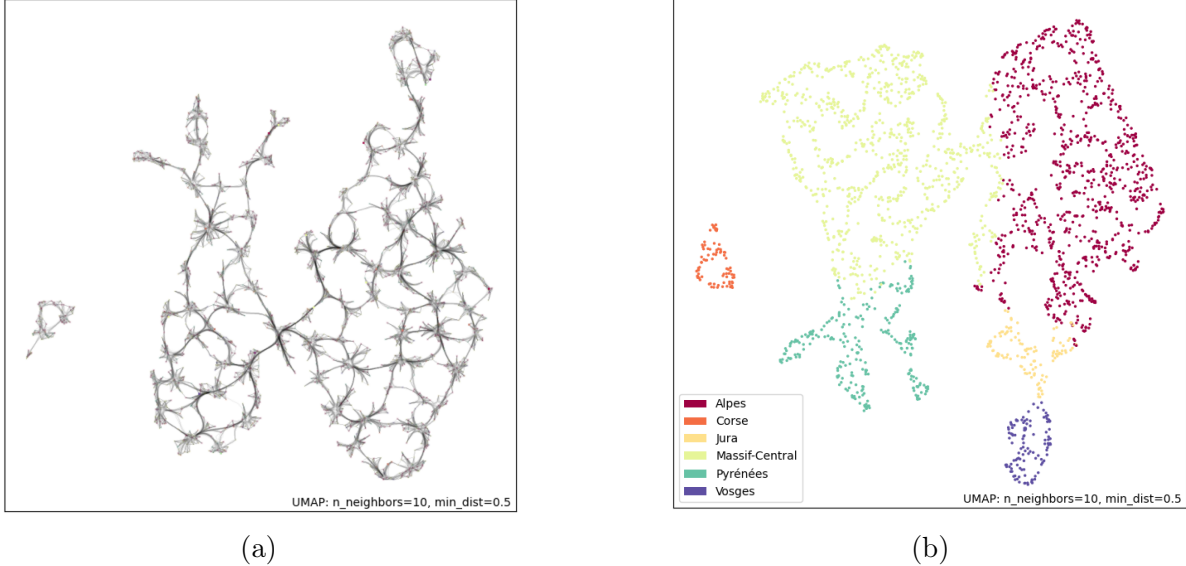


Figure 6: (a) Graph of stations imputed from exogenous predictors with umap algorithm. (b) Positions of nodes in exogenous predictors space after umap dimensionality reduction colored by region.

4.2 Training and evaluation

Training framework: To train the architectures, we find the learnable parameters W that minimize a given average error measure between the output of the neural network $\hat{X}_{[t,t+T]} = f(X_{[t,t+T]}, M_{[t,t+T]}, W)$ and the expected output over a window time period $[t, t + T]$. Here, two main methodological choices are possible.

The first one would be to train the model to output time series similar to the accessible (input) ones in a continuous way anywhere in the dataset as:

$$\mathcal{L}(\mathbf{X}_{[t,t+T]}, \hat{\mathbf{X}}_{[t,t+T]}, \overline{\mathbf{M}}_{[t,t+T]}) = \sum_{n=t}^{t+T} \sum_{i=1}^d \frac{\langle \overline{\mathbf{m}}_t^i, \ell(\mathbf{x}_t^i, \hat{\mathbf{x}}_t^i) \rangle}{\langle \overline{\mathbf{m}}_n^i, \mathbf{m}_n^i \rangle}, \quad (22)$$

with $l(\cdot, \cdot)$ the error measure and where $\overline{M}_{[t,t+T]}$ corresponds to the binary complement of $M_{[t,t+T]}$. The error measure is averaged over the d stations of the dataset and time window $[t, t + T]$. This corresponds to an unsupervised setting with the assumption that temperatures generated in data holes will follow the same physics as accessible ones, and be most probably close to ground truth. This way, the model is generative and learns the underlying physics of the time-series signal.

The second one consist in training the network to reconstruct missing data while available data is always taken as output when possible as:

$$\mathcal{L}(\mathbf{Y}_{[t,t+T]}, \hat{\mathbf{X}}_{[t,t+T]}, \mathbf{M}_{[t,t+T]}) = \sum_{n=t}^{t+T} \sum_{i=1}^d \frac{\langle \mathbf{m}_t^i, \ell(\mathbf{y}_t^i, \hat{\mathbf{x}}_t^i) \rangle}{\langle \mathbf{m}_n^i, \mathbf{m}_n^i \rangle}. \quad (23)$$

This way the model learns the imputation task itself, to complete missing values when there is holes of data, coherently with the fact that available data is an absolute truth, in addition to the underlying physics needed to generate meaningful time series.

Learning the physical structure in an unsupervised way may be more fundamental and generalizable. In particular, It should be possible to train the model at imputation out of sample (complete missing values in a different window than the one given as input) that can then be executed in small spatial windows, scaling better technically with the size of the dataset as it would be independent of specific stations (every station corresponds otherwise to a spatial dimension). For these reasons, it would be an interesting approach to explore; but training with this method is harder and gives worse performances in our case. With the clear task at sight of completing a given dataset, we pursue with the second in sample method, specializing to the structure of the given stations. To do this in a self-supervised way, we then need to add a second artificial masking on which to train the model at the imputation task. Learning the imputation task itself comes with default of specializing to encountered missing patterns, such as proportion of missing data. To work around this problem, we sample the artificial mask from different missing probabilities along training of the neural network, while normalizing the error measure by the number of missing values to give the same importance to every missing pattern at every training step (see C.3).

As usual in machine learning, we split the dataset into train test and validation subsets. With time series data modality, one cannot randomize this split because of time coherence. We then fix

the subset sizes before splitting in three timewise, with the training subset over the most recent period (with the most available data) to be coherent with the final reconstruction task. Batches of data (collections of vectors of multivariate time series over different windows) are then randomly sampled within these subsets and fed to the architectures. Weights of the networks are optimized by minimizing the loss (the error measure over which we train) with stochastic gradient descent and cosine annealed learning rate until convergence (for detailed training settings see [B](#)). Performances of the model are then evaluated on different datasets. Hyperparameters of the architectures are fixed on the validation dataset. Performances of trained architectures with fixed hyperparameters are then evaluated on unseen data on the test dataset.

Error measures: To evaluate the models performances we make use of classical statistical losses, distances between predicted and ground truth values at every point of multivariate time series. For interpretability and comparability, we define all error measures to have the dimension of temperature and units of celsius degrees. Basic metrics are then the mean absolute error (MAE):

$$\ell^{\text{MAE}}(y_t^i, \hat{x}_t^i) = |y_t^i - \hat{x}_t^i|, \quad (24)$$

for simple performance evaluation, and the mean squared error (MSE):

$$\ell^{\text{MSE}}(y_t^i, \hat{x}_t^i) = (y_t^i - \hat{x}_t^i)^2 \quad (25)$$

to penalize large errors potentially induced by a bad representation of the way to solve the task.

Even if the above metrics allow learning of the task fundamentally without biases, our objectives for the architectures are more specifically defined. As the output is intended to be input of a physical model, generated time series have to be physically coherent. Although for example, MSE and MAE would not penalize noisy time series with the ground truth as average, even with large variance. Moreover, big temperature discontinuities would not necessarily give rise to large measured errors if they are rare enough over the dataset. Particularly, large (non physical) errors made on statistically less represented stations would not be penalized, and probably even benefit the learning of a simpler task valid on typical and more probable stations leading to wrongly lower measured errors. For this reason, we need to introduce error measures of physical coherence for the evaluation of models.

A simple physics informed metric to introduce, enforcing continuity, is the sum of gradients (to be minimized). It can be defined with an overall weight depending on the physically acceptable size of jumps of temperature, that depend on the discretization precision of the data. This metric particularly makes sense temporally in our case, as continuity surely has to be respected between hourly steps of temperature. It will have a diffusion effect on the predicted times series, and again, we can take the squared gradients to penalize more large non physical jumps. We note it as:

$$\ell_{\partial t}^{\text{MAE}}(\hat{x}_t^i) = \left| \frac{\partial \hat{x}_t^i}{\partial t} \right|, \quad \ell_{\partial t}^{\text{MSE}}(\hat{x}_t^i) = \left(\frac{\partial \hat{x}_t^i}{\partial t} \right)^2. \quad (26)$$

On the other hand, spatially, the discretization of the underlying temperature field by the observation stations set is coarse: geographical distances between stations are larger than the typical variance scale of individual station temperature changes. But at larger scales, coherent temperature structures can exist, induced for example by shared topography (multiple stations in the same

valley) and in general by meteorological phenomenon (large scale advection of air masses sharing relatively common origins and properties). These meteorological phenomenon can take place at different spatial (and temporal) scales. One then needs to introduce a measure of multiscale physical coherence. As different temporal scales of first order weather phenomenon are very well defined (diurnal and annual cycles), we focus on applying this measure to spatial information and rely on continuity for the physical coherence of temporal structure.

To construct this error, we apply iteratively the following procedure: identify neighborhoods of nodes (stations) on the graph, average temperature within every clustered neighborhood, and assign the resulting temperatures to all nodes pertaining to each given cluster. By doing so, small scales (fast degrees of freedom) are integrated over at each step and the resulting graph is renormalized to the base scale. Every step then highlights the temperature structure at a different scale (Fig.7). By applying this procedure to both original and reconstructed data, we can evaluate the multiscale structure coherence by summing their differences at every step as:

$$\ell_{\text{RG}}^{\text{MAE}}(y_t^i, \hat{x}_t^i, A^{ij}) = \sum_n w_n \sum_{\substack{\{i|\hat{x}_t^i \in \text{KNN}(A^{ij}, n)\} \\ \{j|y_t^j \in \text{KNN}(A^{ij}, n)\}}}^n \frac{|\hat{x}_t^i - y_t^j|}{n}, \quad (27)$$

where A^{ij} is the adjacency matrix encoding for the underlying graph and n denotes the number of nodes per neighborhood at every step (decreasing as the accessed scale grows). With the parameters w_n , the measure can be weighted at different scales to account for the known physical scales of the meteorological phenomenon present in the data. KNN (K-nearest neighbors) denotes the clustering algorithm used to find constant number or neighbors at every step of the process (contrarily to a lattice, there are no regular nearest neighbors in a graph). Again, to penalize strongly non physical signals we can square the difference, noted $\ell_{\text{RG}}^{\text{MSE}}(y_t^i, \hat{x}_t^i, A^{ij})$. In the following we call this error measure the renormalization group error, by the name of the statistical physics procedure from which it is inspired.

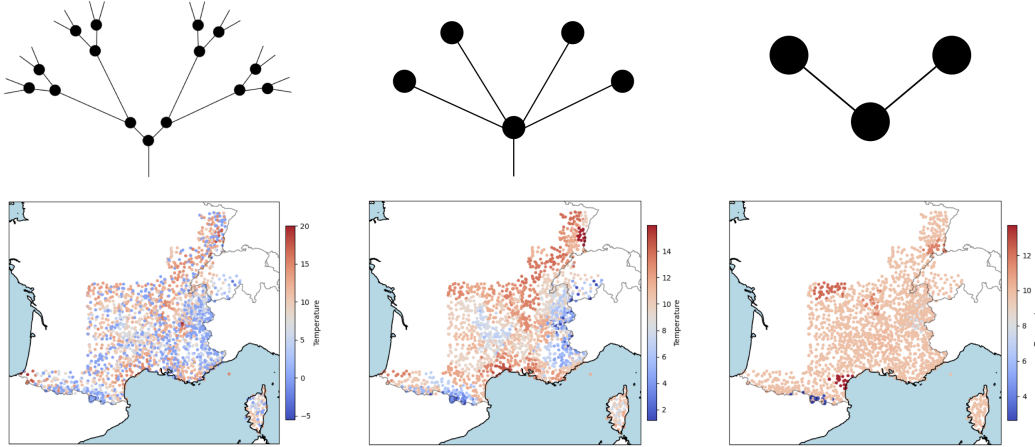


Figure 7: Representation of different steps of the renormalization group error measure procedure applied to the temperatures dataset.

All the above error measures can be used both as evaluation metrics and as losses (learned to be minimized by the neural network along training). Although in practice, computing the renormalization group error at every step of training is expensive. In the following, we then trained our architectures on a composite MSE and temporal gradient loss:

$$\begin{cases} \mathcal{L}_{\text{train}}(\mathbf{Y}_{[t,t+T]}, \hat{\mathbf{X}}_{[t,t+T]}, \mathbf{M}_{[t,t+T]}) = \left(\sum_{n=t}^{t+T} \sum_{i=1}^d \frac{\langle \mathbf{m}_t^i, \ell(\mathbf{y}_t^i, \hat{\mathbf{x}}_t^i) \rangle}{\langle \mathbf{m}_n^i, \mathbf{m}_n^i \rangle} \right)^{\frac{1}{2}} \\ \ell(\mathbf{y}_t^i, \hat{\mathbf{x}}_t^i) = (y_t^i - \hat{x}_t^i)^2 + \alpha T \left(\frac{\partial \hat{x}_t^i}{\partial t} \right)^2 \end{cases} \quad (28)$$

where α is the relative weight of gradient loss and the square root and factor T are taken to get dimensions of temperature. The multiscale spatial coherence of reconstructed signals with different methods is evaluated afterwards.

Reconstruction tasks: To evaluate the methods we consider two tasks with different purposes. We collect ground truth data from the original dataset from the largest dense subset with the least missing values as possible (by selecting time window and dropping too corrupted stations). On a two years dataset with approximately 500 stations we consider a homogeneous block missing random corruption pattern with different missingness probabilities (Fig.8a). On this dataset we evaluate the fundamental performances of the method at a simplified reconstruction task. It is meant to validate the framework, run machine learning experiments, and set all methodological choices.

But before actually reconstructing the full dataset (on which it is not possible to evaluate performances), it is interesting to study the behavior of the method on a similar setting. To reproduce artificially the asymmetry of missing values of the historical data reconstruction task, we sample a mask from the original distributions of missing values. To do so, we sample times of first record (for monotone missing pattern) and probability of missing with typical block size (for random missing pattern) from kernel density estimation fitted to the corresponding distributions on the original dataset. We obtain a dataset of approximately 200 stations over 30 years with the same corruption structure as the original one (Fig.8b). On this dataset we characterize the behavior the method on extreme corruption patterns and study the climatological properties of the reconstructed time series to get a physical evaluation of the method.

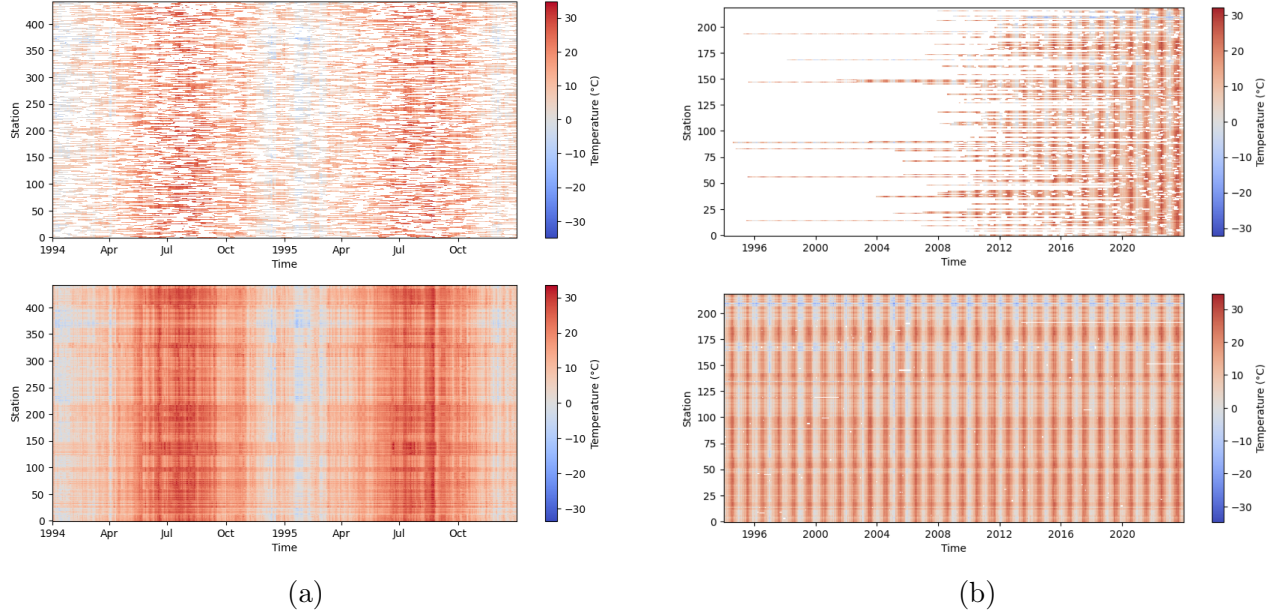


Figure 8: (top) Artificially corrupted and (bottom) ground truth datasets for (a) the homogeneous reconstruction task and (b) the heterogeneous (historical data) reconstruction task.

5 Results and discussion

5.1 Models performances (homogeneous task)

In this section we analyze the performance of the different statistical and machine-learning models from section 4.1 for the homogeneous reconstruction task, via the classic and more specific metrics presented in section 4.2. The results are displayed in Tab.1. As discussed before, at the spatial extent of the dataset most stations share the same temporal structure, so that when averaged over the spatial dimension mean imputation methods already achieve limited errors, with an MAE lower than 3°C . On the other hand, even if obtained MSE and MAE could be acceptable from a climatological perspective, the generated time series are non physical. As expected from a simple model, the generated signals are oversimplified and smoothed, with large jumps and large errors on statistically rare stations (see D). This translates into large errors in physical losses (Tab.1) and would give rise to non climatologically coherent signals. Introducing a physical model as prior such as in GC mean enforces more structure which translates into lower errors, both in the standard and more physical metrics. A further step to improve the GC mean could be to specify the statistical method to what is physically known from the data (for example imposing the importance of altitudinal gradient in a mountain stations dataset).

All error metrics are distinctively lower for neural network based methods (Tab.1), that are able to learn both the underlying physical structure and the completion task itself, not only achieving better performances but enforcing continuous imputation at the limits of the data holes. The MLP learns from the spatial structure at each timestep, but without particular architecture structure as prior. Because of its better representation of information, spatially with the graph structure and temporally with memory networks, the GRIN architecture achieves the best performances,

approaching 1°C of average imputation error on a dataset with 90% of missing values (Tab.1).

Table 1: Reconstruction performances on homogeneous masking with 90% missing value

models	reconstruction error (°C)			
	RMSE	MAE	$\mathcal{L}_{RG}^{\text{RMSE}}$	$\mathcal{L}_{RG}^{\text{MAE}}$
mean	3.52 ± 0.17	2.80 ± 0.12	1.25 ± 0.11	0.92 ± 0.06
GC mean	2.93 ± 0.08	2.30 ± 0.06	0.95 ± 0.05	0.69 ± 0.03
MLP	2.26 ± 0.06	1.78 ± 0.04	0.67 ± 0.03	0.49 ± 0.02
GRIN	1.61 ± 0.04	1.18 ± 0.02	0.25 ± 0.00	0.17 ± 0.00

Beyond errors, GRIN reconstructs physically coherent time series; both in summer where clear diurnal cycle helps and in winter with less periodic patterns (Fig.9a). On More peculiar stations like high altitude ones, the model does perform worse, but still reconstructs physical time series, potentially with bias ore even carrying diurnal patterns into noisy regimes of small variance (from constantly cold station in winter -Fig.9b). This physical coherence is reflected into the physical losses. But we note that, even with 90% of missing values, the alternating presence of different observations stations along the time series does not correspond to the hardest reconstruction regime in the dataset. In particular, it is not equivalent to the historical data reconstruction regime where the same stations are missing over very long periods.

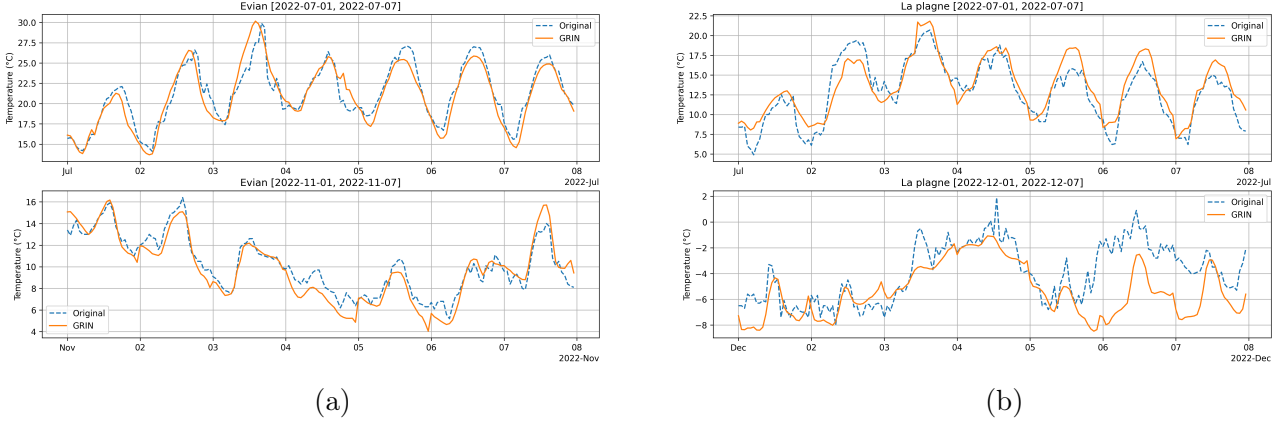


Figure 9: Example of (top) summer and (bottom) winter time series at stations (a) Evian (alt. 395m) and (b) la Plagne (alt. 2120m) reconstructed by GRIN (for more examples see D).

5.2 Reconstruction of historical data (heterogeneous task)

Here we examine more specifically the performances for the inhomogeneous reconstruction task. To take into account the difficulty of an asymmetric reconstruction and hard regime (similar absent stations over long time periods, monotonous missing pattern), we evaluate the method on the second reconstruction task over climatological periods (Tab.2). In a similar way to what has to be done for the final task, the architectures are trained only on the recent period (2014-2024 in Fig.8b), and then used to reconstruct the signal backward in time. It can be noted that this artificial smaller

scale task is probably a harder setting than the one of the complete reconstruction, as it leaves less data for training and a dataset ten times sparser than the complete one in the densest periods (making it harder to recover spatial structure). For this heterogeneous reconstruction task, all metrics are degraded with respect to the homogeneous task, in line with its increased complexity. Again, the graph neural network architecture (GRIN) performs better than the statistical and neural network baselines. Hourly temperature time series are reconstructed with an average error of 1.5 to 2°C (while the mean and GC mean imputations reach or overpass 4°C in RMSE) and performant physical losses, in a regime where more than 90% of the data is missing, twenty years before the training period (Tab.2).

Table 2: Reconstruction performances on heterogeneous masking fitted on original dataset

models	reconstruction error (°C)			
	RMSE	MAE	$\mathcal{L}_{RG}^{\text{RMSE}}$	$\mathcal{L}_{RG}^{\text{MAE}}$
mean	4.41 ± 0.49	3.31 ± 0.28	1.92 ± 0.91	1.46 ± 0.70
GC mean	3.97 ± 0.65	2.89 ± 0.35	1.77 ± 0.97	1.34 ± 0.72
MLP	2.52 ± 0.79	1.96 ± 0.51	1.13 ± 0.68	0.91 ± 0.50
GRIN	2.05 ± 0.40	1.55 ± 0.24	0.79 ± 0.32	0.60 ± 0.22

The GRIN architecture is able to predict time series with more physical structure, in a clearer way in this hard regime. This is typically illustrated in Fig.10a where we can recognize the spatial structure of the dataset in bands of cold stations present in the recent periods. Backward in time along the reconstruction by MLP, these structures shade into a smoothed signal, accompanied by the augmentation of average window errors. Because the spatial structure is enforced by the graph and propagated by the temporal memory in the GRIN architecture, these structures are present along all the reconstruction made by GRIN (Fig.10b). In fact, we use overlaps between every prediction window to take advantage of past reconstructed information as a guide for the imputation of the next steps. Even if the error has to follow the same pattern as for MLP or other reconstructions (worse performances as data availability scatters), GRIN’s performances are again reflected into the maintain of lower physical errors. Although, to study more precisely how much the GRIN architectures allows mitigating this error growth, it would be interesting to set a higher statistical bound to which suboptimal methods would probably saturate.

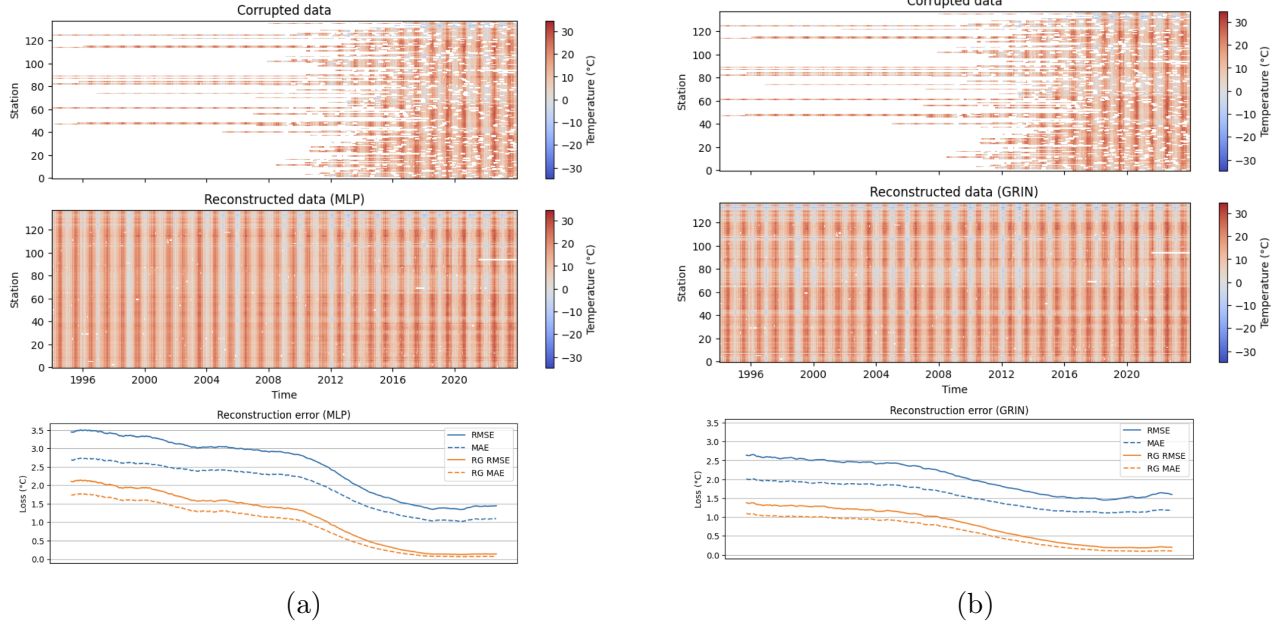


Figure 10: Reconstruction of asymmetrically missing historical data with associated reconstruction errors by prediction window for (a) MLP and (b) GRIN architectures.

5.3 Climatological analysis

With access to these long term reconstructions, we can conduct climatological analyses of reconstructed time series. Temporal trends in temperature signals are of particular interest for the assessment of climate change in mountain regions; hence we conduct a comparison between the computed trends (over 30 years of data) from the reconstructed dataset, and the non corrupted one (Fig.11a). As a first check, station average temperatures reconstructed with GRIN are strongly correlated with original ones, without dependence of error with temperature or altitude (Fig.11a). The mean annual temperature change of reconstructed time series is correct in average, pointing towards the global warming signal with $+0.05^{\circ}\text{C}/\text{yr}$ found in the literature for this mountain region (e.g. Monteiro and Morin, 2023). However, station-wise, the trends of the reconstructed temperatures exhibit a large variance (Fig.11b). As expected intuitively, this variability seems to be correlated with altitude. In this direction, more meteorological analysis of the architectures reconstruction, and their modification to fit domain knowledge accordingly, are the next steps of development of the method.

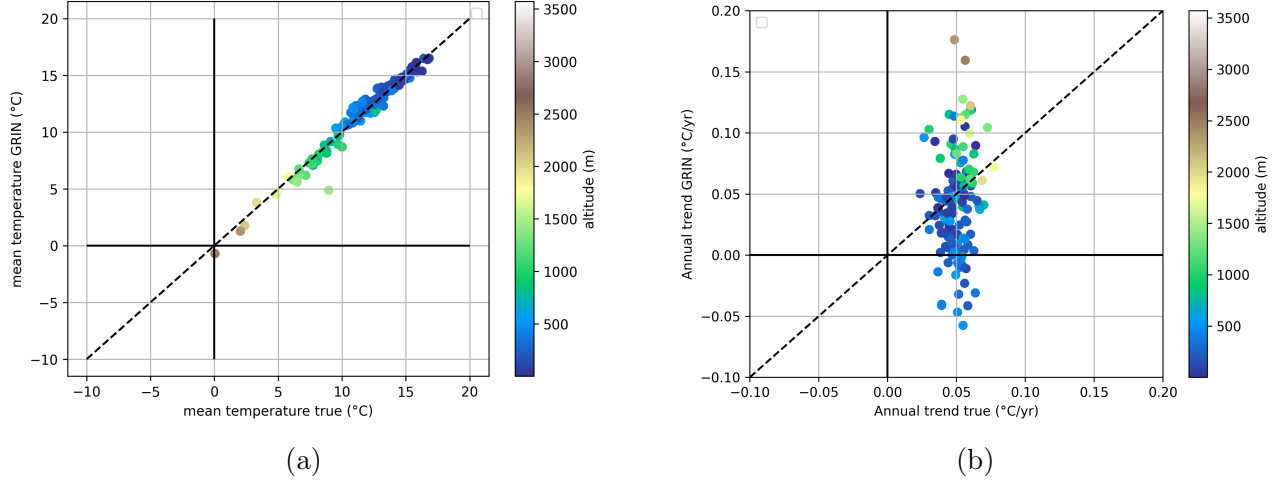


Figure 11: (a) Mean station temperature of reconstructed dataset against non corrupted one. (b) Mean annual temperature trend computed from reconstructed dataset against non corrupted one.

6 Conclusion and perspectives

In this project, we developed a framework for the reconstruction of historical meteorological time series with the Graph Recurrent Imputation Network (GRIN) architecture. Confronting it to both statistical and machine learning baselines, we evaluated and fixed the training and architectures methodological settings to the specificities of the targeted task. The performances and behavior of the method have been evaluated on two representative tasks: a homogeneous reconstruction, setting the architectures hyperparameters and quantifying fundamental performances; and a heterogeneous historical data reconstruction characterizing the behavior of the method in a similar setting to the one of the final objective task. To reconstruct physically coherent time series, we introduced temporal and spatial multi-scale physics-informed error measures. Finally, we started to study the climatological consistence of the reconstructed signals.

Even if they cover both statistical and learning approaches, the mean imputation and multi layer perceptron methods only bear the role of baselines; expected to be outperformed by any more complex architecture that would work. These baselines then allowed to validate the good performances of the GRIN method, but not its optimality. Before actually using the method, one would need to confirm the usefulness and good working of every complexity layer (every introduced prior) of GRIN, to understand what the architecture really learns. A good way to do it would be to study the performances of simpler architectures constituting building blocks of GRIN, with the same philosophy to what is done in ablation studies. Among the next steps of the project would then be the implementation of a RNN alone (to ablate the graph structure), and of a transformer alone (to ablate the temporal memory). It would even be possible that the intentional characteristics introduced with GRIN are not all suited to express the underlying processes in the dataset, and that an architecture with fewer layers of complexity perform better.

An other important next step for the project would be to develop further the meteorological study of the signals generated by GRIN, to understand what it learns from a physical perspective, and potentially improve the architecture accordingly. Once performances are satisfactory, the

method can be used to reconstruct the actual dataset of historical near-surface air temperatures in the French mountain regions, as input to the S2M reanalysis. But by definition, it will not be possible to measure achieved performances for this reconstruction; and one needs to conclude on what is considered satisfactory. At this point, it is important to note that temperatures reconstructed with this method do not constitute an observational dataset, but the outputs of a model with its own performances and biases. The objective of this work was to specialize the method to the reconstruction of missing values, targeting a different problem than learning the purely physical processes involved in the atmospheric model used in the reanalysis methodology. Epistemologically, this model is to be seen as a small data processing reanalysis: the architecture is made to impute data holes, while a rough physical model of first order processes is learned from data by the neural network to generate physically consistent signals.

In a longer-term perspective, after the reconstruction of the temperatures dataset, it will still be possible to improve the method in multiple ways. As an example, by construction, it is already possible to process vector time series with GRIN, and to reconstruct multiple meteorological fields (pressure, precipitations...) in parallel, possibly leveraging their inter-correlations for better performances (e.g. low pressure and precipitations positively correlated with low temperatures/high humidities).

To conclude, compared to other deep learning architectures (learning abstract internal representations freely and from data), GRIN as used in this project is relatively small, involving a limited number of trainable parameters. But as it happens, the meteorological data (and the underlying processes) is very low dimensional (lie in a low dimensions manifold), represented by very structured scalar values, well described by defined sets of equations. We encounter the symptoms of these characteristics when performing machine learning experiments, where theoretical behaviors normally encountered in ideal settings appear clearly on the meteorological data (see [C.1](#)). This is a advantageous in multiple ways. First, it enables a scientific treatment of the problem from the machine learning perspective, making use of results obtained from theoretical machine learning to fix the architectures parameters without only (uninformed) cross validation optimization. Moreover, it offers a great setting for machine learning sobriety, without too many experiments or too big models, and with relatively low compute requirement.

With these final remarks, we orient this project toward its final perspective, an actual implementation of the method for the reconstruction of historical temperature records at Météo-France, in support of the elaboration of reanalysis products. Deliverables of this work are, with this report, a python implementation of the full method, with user guides and comments, and flexibility for further improvement.

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A Reproducibility

All the code needed for the implementation of the method is made available on [GitHub](#).

B Detailed training settings

In this appendix, we give more details on the experimental settings used to train our models. We train all the models by randomly sampling batches of 15 elements per training step. Batches of data can be purely spatial at one time step or spatio-temporal with window size corresponding to the prediction size of the model (Fig.12). Models are trained over 250 epochs for final results and 100 epochs for machine learning experiments; with stochastic gradient descent and a cosine annealing scheduler for learning rate.

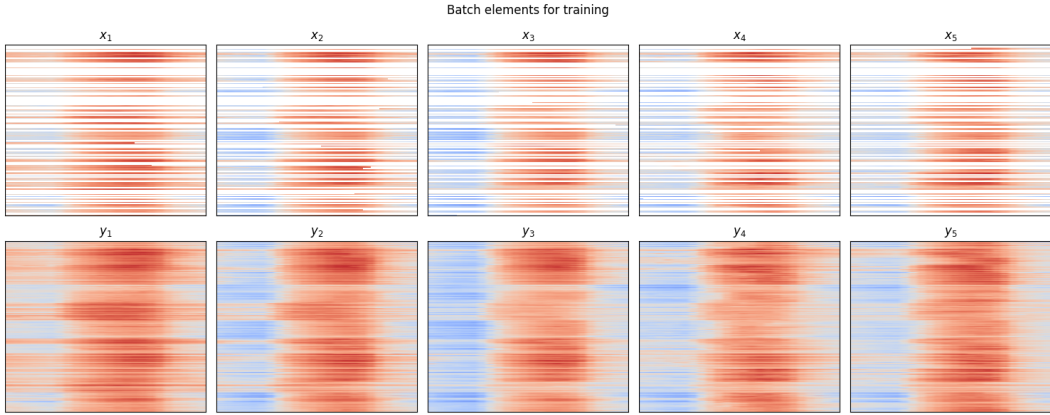


Figure 12: Example of spatiotemporal batches for training.

For training, datasets are split into train and test along time dimension with sizes $N_{\text{train}} = 0.8N$ and $N_{\text{test}} = 0.2N$, with train subset selected over the most recent period. For the asymmetric reconstruction task, the training dataset is restricted to the recent period with enough data (2014-2024) in the same way that it would be done for the real task. The inference phase reconstructing the dataset is made separately, with the trained network sweeping over timeseries backward window by window, possibly with an overlap stride.

Hidden dimensions for spatial latent representations of all networks (MLP, GRIN spatial decoders, GRIN MLP combining forward and backward blocks) are fixed to 100 for the homogeneous reconstruction (approximately 400 stations) and 64 for the heterogeneous reconstruction (approximately 120 stations) for all executed experiments. In a general way, the best hidden dimension for spatial information is found to be inferior and of the order of magnitude of the number of stations (see C.1). The hidden dimension for temporal latent representation of all networks is fixed to 24. For networks with temporal coherence, the best prediction window is found to be 24 (hours); consistent with the fact that it automatically introduces diurnal temporal structure in the architectures. Corresponding training batches and inference windows are then 24 hours too.

C Machine learning experiments

C.1 Size of latent spaces

To fix the size of latent space (hidden dimension) for all networks, we search for the size where the architecture becomes too complex compared to the data and starts learning the noise, worsening in performances. This phenomenon is called overfitting and is located when the test error grows while the train error continues approaching zero. An architecture of optimal size and complexity is a network that is large enough to learn only the generalizable representations from the dataset. Coherent with the importance of spatial information in the studied dataset (compared to temporal information that is already very structured), this optimal hidden size is found to scale at first order with the number of stations, being inferior to and of the same order of the spatial dimension of the dataset (Fig.13, Fig.14). This principle is then applied in this work to all the choices of hyperparameter selection for neural network architectures.

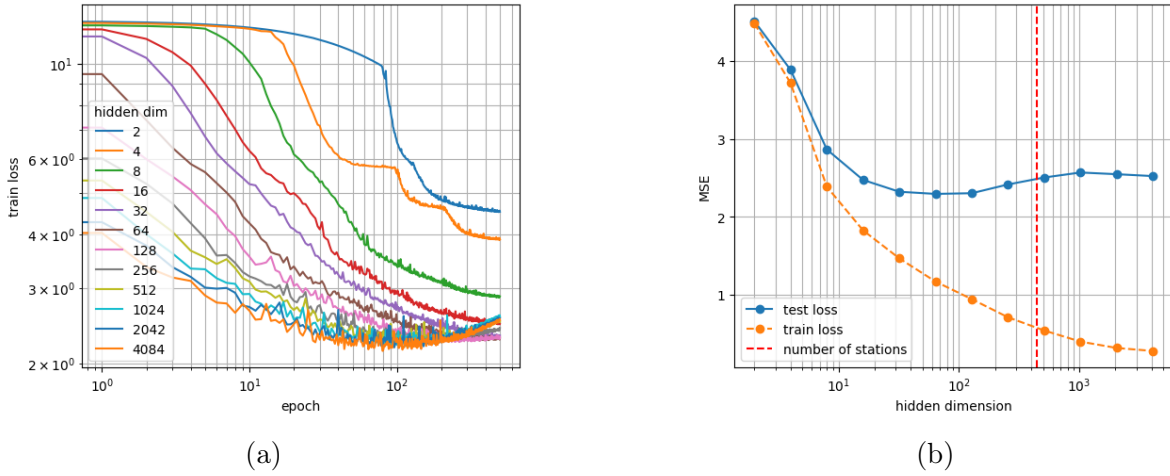


Figure 13: (a) Training losses and (b) final training and test errors for one layer MLP with different hidden dimensions.

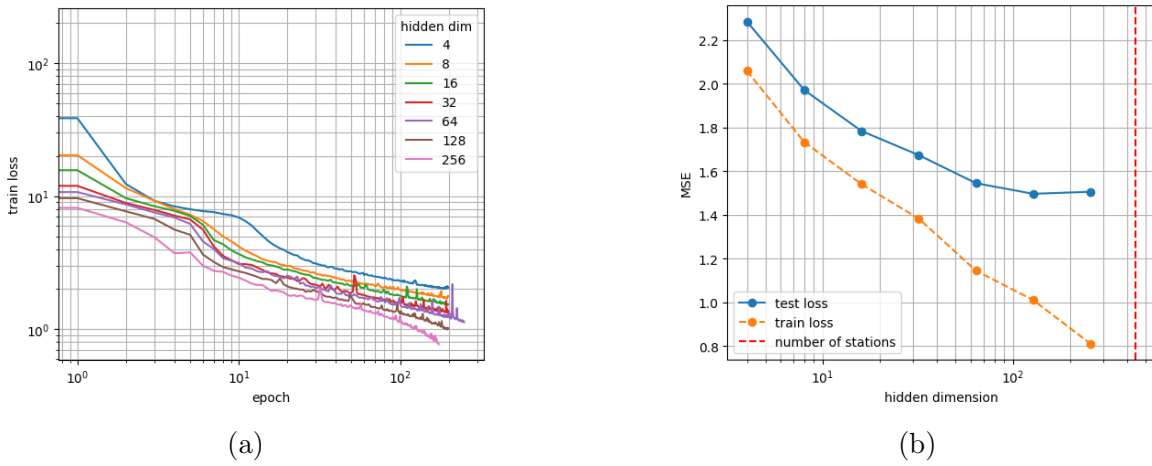


Figure 14: (a) Training losses and (b) final training and test errors for one layer GRIN with different spatial hidden dimensions.

It is interesting to note that the theoretically known behaviors of neural networks appears clearly with our dataset. In Fig.13b we recognize the double descent phenomenon with the interpolation point (local maximum test error and zero train error) corresponding to the number of degrees of freedom present in the dataset (larger than its spatial dimension). An other example are the plateaus followed by large drops in train errors of networks with very small hidden dimensions (autoencoders) (Fig.13a); corresponding to the specialization of each hidden neurons to different features in the data along training. The number of possible phase transitions (drops in error) then corresponds to the number of hidden neurons in the neural network; the corresponding multiple drops are smoothed (by finite size effects) into a continuous decrease when this number is large. These clear behaviors of networks on meteorological data are interesting to make informed machine learning methodological choices.

C.2 Graphs

The underlying graphs for the GRIN architecture can be constructed in different ways. Directly from data: we can pick as edges weights the values of the spatial correlation matrix. From exogenous predictors (stations metadata): we can cluster the stations by distances in the predictors space. This connects nodes corresponding to similar typologies of stations. This can be done efficiently with KNN (K-nearest neighbors) algorithm, or in a less straightforward way leveraging non linear correlations with umap algorithm. Graphs inferred from correlation matrix exhibits less pronounced structure (as it can be seen in the adjacency matrix) accompanied by higher errors (Fig.15). The use of predictors is then a clear advantage. As different clustering algorithms give similar results, the simplest one (KNN) is the one used in this work. The graph structure can be recovered from the correlation matrix by imposing a large threshold for the assignment of an edge to the graph (connecting only very correlated stations). The recovered structure is to some extent similar to the one inferred by the clustering algorithms, but the method never reaches the performances obtained when using predictors (Fig.16).

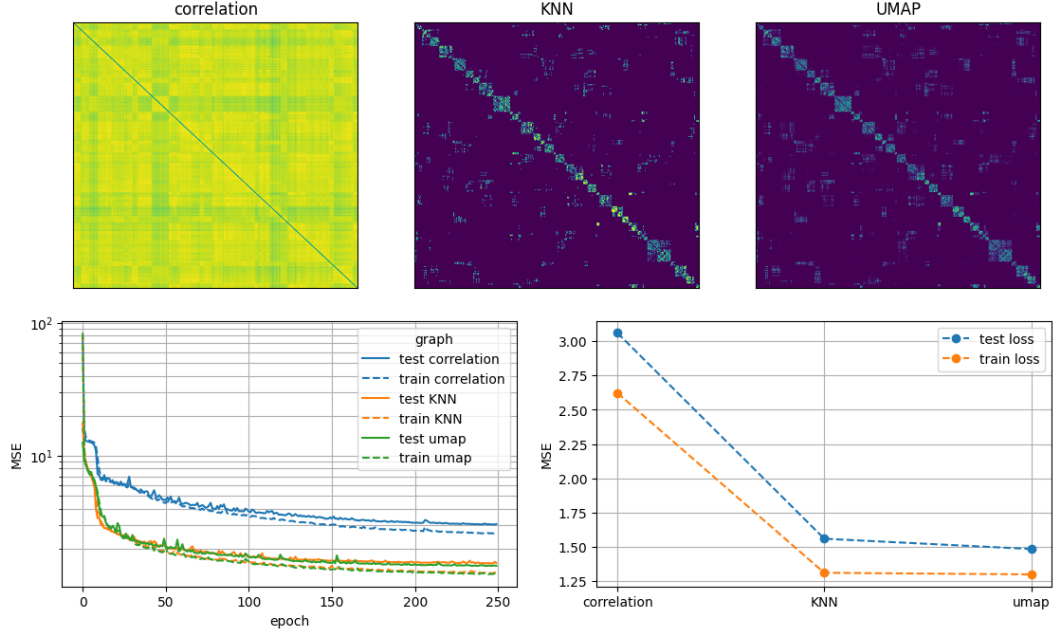


Figure 15: (top) Adjacency matrices of the graphs inferred from the correlation matrix and from predictors with KNN and umap algorithms. (bottom left) training curves and (bottom right) final errors obtained for GRIN with different graphs. Values of the weights connecting all pairs of nodes in the graph (between 0 and 1) are color-coded from blue to yellow in the adjacency matrices.

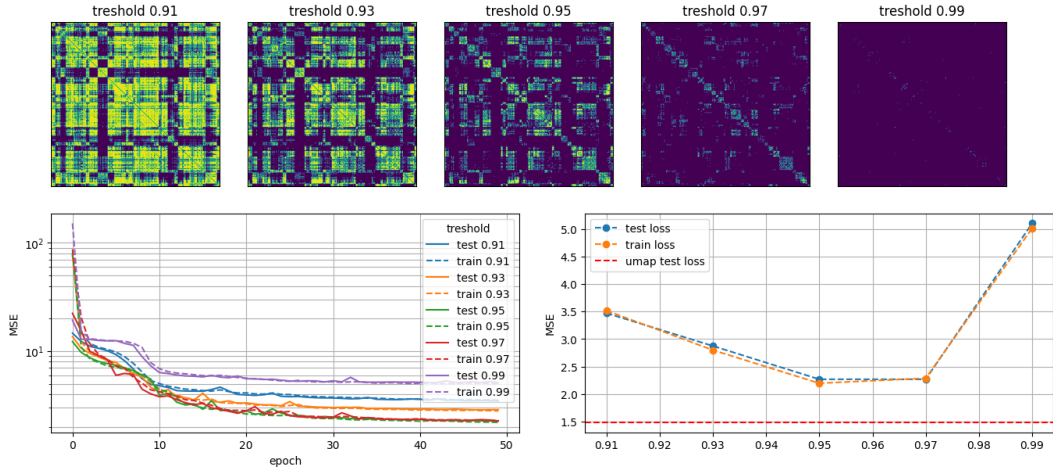


Figure 16: (top) Adjacency matrices of the graphs inferred from the correlation matrix with different minimum threshold for edge assignment. (bottom left) training curves and (bottom right) final errors obtained for GRIN with different thresholds. Values of the weights connecting all pairs of nodes in the graph (between 0 and 1) are color-coded from blue to yellow in the adjacency matrices.

C.3 Robustness to masking

When training in a self-supervised way adding a second layer of corruption mask to create training samples, the architecture specializes to solve the task with most encountered corruption processes. As we see in Fig.17, an architecture trained on masking with a certain probability performs worse when reconstructing datasets with other corruption probabilities. The best performance is always on the dataset encountered in training, and results suggest that it is penalizing to train on too corrupted datasets. To work around this problem, we sample the artificial mask from different missing probabilities along training of the neural network, while normalizing the error measure by the number of missing values to give the same importance to every missing pattern at every training step. Models trained in this way (abusively called diffusion) not only generalize better to multiple missing probabilities, but perform even better than specialized architectures on their own training patterns (Fig.17). This suggests that architectures trained on different types of masking learn a more fundamental and correct representation of the imputation task.

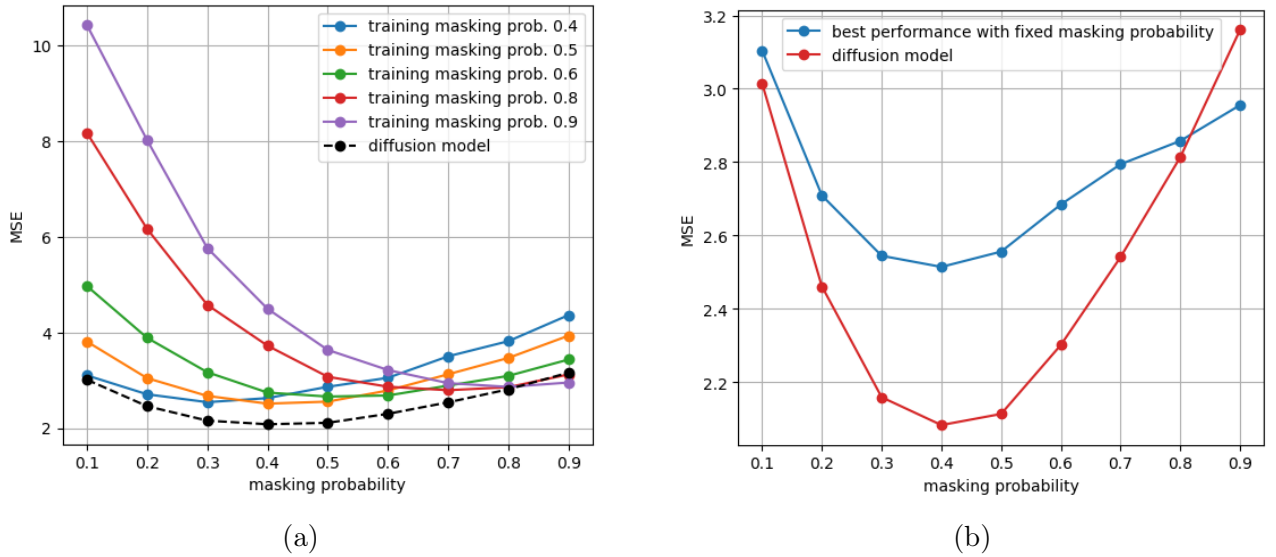
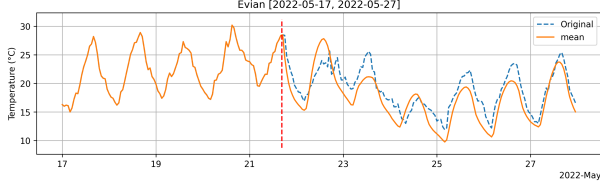


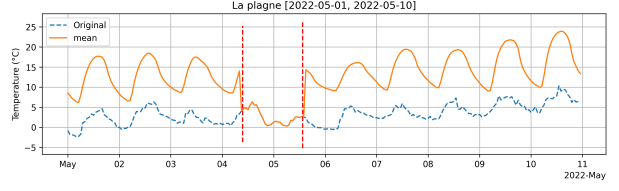
Figure 17: (a) Performances of models trained on specific masking probabilities at reconstructing datasets with different masking probabilities. (b) Best available performance among specialized models against (diffusion) model trained on multiple corruption patterns.

D Supplementary results

As discussed in the main results, simple models as mean imputation bears a too simple representation of the task; performing correctly on well structures statistically usual (low altitude) stations (Fig.18a), but giving large errors on rare (high altitude) ones. As these statistical methods don't learn the imputation task itself, these large deviations give rise to non physical time series accompanied by large temperature jumps (Fig.18b). In comparison, neural network MLP architectures learning the imputation task generates smooth time series signals, even at the boundaries of data holes and on statistically rare stations (Fig.19). Although, because of their weaker physical expressiveness compared to GRIN, they still tend to perform worse on more peculiar (high altitude winter) time series signals (Fig.20).

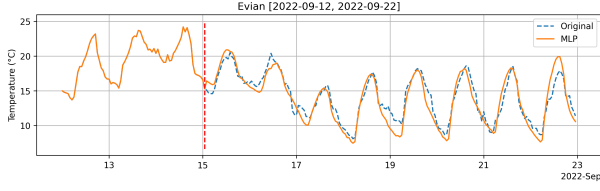


(a)

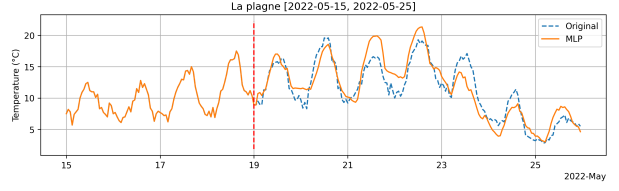


(b)

Figure 18: Example of time series at stations (a) Evian (alt. 395m) and (b) la Plagne (alt. 2120m) reconstructed by mean imputation. Dashed red line indicates limits of a data hole.



(a)



(b)

Figure 19: Example of time series at stations (a) Evian (alt. 395m) and (b) la Plagne (alt. 2120m) reconstructed by MLP. Dashed red line indicates limits of a data hole.

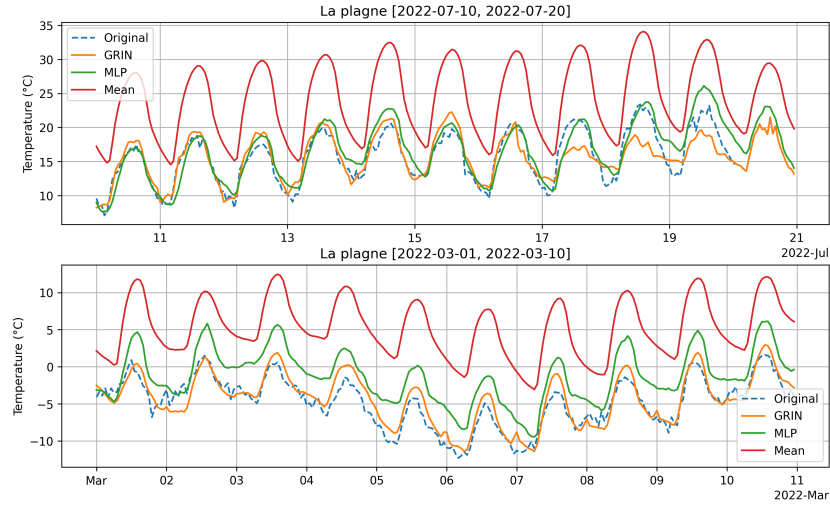


Figure 20: Reconstructed time series of high altitude station (la Plagne, alt. 2120m) obtained with mean imputation, MLP, and GRIN methods.